Instituto Politécnico Nacional
Centro de Investigación en Computación

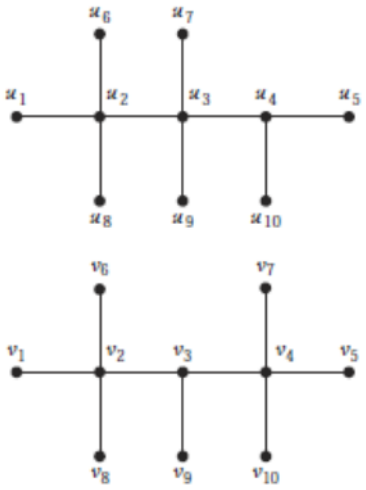
Mathematics for Computer Science

Nombre Completo: Mondragón Vega Fernando Iván
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Homework 6

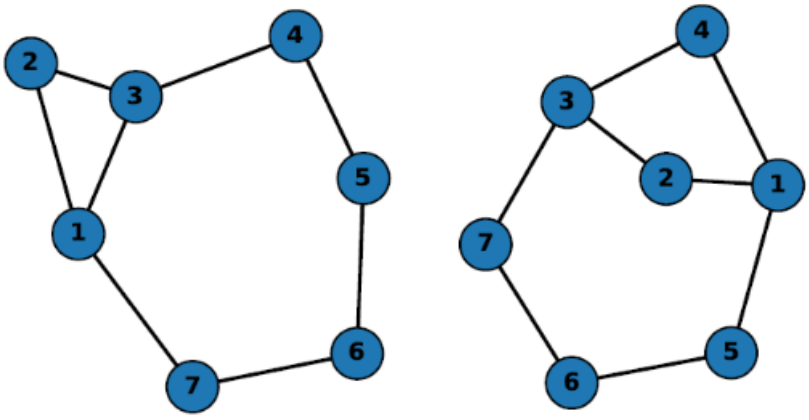
Determine whether the following graphs are isomorphic or not, and explain why.

1.



Solution
These graphs are not isomorphic. The reason is that the one on top has two vertices, $u2$ and $u3$, of degree 4 that are adjacent. This is not the case for the two vertices with degree 4 in the bottom graph $(v2, v4)$.

2.



Solution
The graph on the left has two nodes with degree 3 that are adjacent. The graph on the right also has two nodes with degree 3, however they are not adjacent. Therefore, these graphs are not isomorphic.